

Dhrystone MIPS (DMIPS): The unit for performance measurement

Dhrystone MIPS is a synthetic computing benchmark program, used to gauge the performance of a MCU. A general 8-bit/16-bit MCU may be in the 5-20DMIPS range, whereas a 32-bit MCU may be in the 40-165DMIPS range. Here MIPS stands for 'million instructions per second.'

MCU of \$2 range," informs Seng.

32-bit architectures

There are several proprietary 32-bit architectures such as Freescale's ColdFire, Renesas' SuperH, Atmel's AVR, Microchip's PIC and Texas Instrument's C2000. However, ARM dominates the 32-bit space with its Cortex-M series architecture licenced to different companies. Some companies like Atmel and TI have their own proprietary architecture as well as microcontrollers based on ARM.

Amongst several known architectures, ARM's 32-bit MCU will become the leading architecture in 2011, ahead of Power Architecture and x86, according to Semacast Research. "ARM architecture continues to gain momentum in the MCU market, with suppliers such as Atmel, Luminary Micro (now Texas Instruments), NXP Semiconductors and STMicroelectronics all demonstrating Cortex-M processor-based MCUs. NXP is also introducing the first parts based on the Cortex-M0 processor," comments Colin Barnden, principal analyst at Semacast Research and study author in his research on the MCU market.

"ARM Cortex-M3 has become the most popular with a strong support eco-system of about 500 companies providing operating system, firmware, tools and integrated development environments (IDEs)," says Shyam Sadasivan, product manager at ARM. "The common architecture and use of tools also helps in easy migration from one vendor to the other," he adds. ARM recently licenced its latest architecture Cortex-M4 and expects launch of some

MCUs based on Cortex-M4 by the end of the year.

Designer need not worry about tools

As embedded engineers move to the next level of designing, the need for more performance and memory will drive the use of 32-bit MCUs. The problems on this ladder are multi-faceted technical requirements; ease of development, availability of development tools and software, time to market and total cost all need to be balanced. 32-bit is still perceived as a complex system.

Designers are also reluctant to switch to a new architecture, as it incurs cost and time. Another constraint is the initial investment in software and tools. A research analyst from Frost & Sullivan says, "The introduction of ready-to-use hardware and software

Applications like industrial automation tools, medical equipment, point-of-sale terminals, digital weighing scales, printers, motor control and displays have migrated from 8-/16- bit MCUs to the 32-bit platform

development tools at competitive prices can provide seamless migration to higher-performance architectures."

Some of the tools for 8-bit MCU systems, such as the IAR tools, GNU tool, Keil tools and Tasking tools, already support 32-bit devices. If you do not want to invest in tools, you can go for open source software (OSS) tools, which are freely downloadable.

As most programs/applications are coded in 'C' language, this makes porting much easier. "Some tools enable you to create a common project structure among all the MCU family, through the project generator. There are tools such as peripheral driver generator (PDG) that help to create the driver

Companies offering low-cost 32-bit MCUs

Atmel, Actel, Freescale Semiconductors, Infineon, Microchip Technology, NXP Semiconductors, Nuvoton Technology, Renesas Electronics Corporation, STMicroelectronics, Texas Instruments

codes. These tools make the porting facilities much easier," shares Seng.

"A developer using 'C' language for programming needs to remember that all memory/data widths are 32-bit. The rest of the things are taken care by the compilers," says Chandak.

32-bit market growth

Research firm Gartner reports that the growth in microcontrollers is happening in the 32-bit space. In 32-bit area, Japanese microcontroller manufacturers Renesas and NEC together are ranked first in both the 32-bit and overall microcontroller market. Most of the MCU manufacturers are focusing on low-cost 32-bit.

"For India, we have started to see more 32-bit MCU usage in these two years, and we will be launching a series of high-performance MCUs for the market needs," informs Seng.

The complexity in terms of design and features for embedded systems is increasing day-by-day. There is a need for a more powerful, high-performance microcontroller unit to cater to this need.

Need for a single common platform, code reusability, high performance, low power requirement, faster processing, aggregation of devices and reduction of the overall cost greatly supports use of the new architecture of 32-bit MCUs.

Moorthy concludes, "Microcontroller manufacturers who take time to understand what problems embedded designers are trying to solve and advocate alternatives that are best to solve the designers' needs are the suppliers that will be sought out over the long term." ●

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